

Product Laws of ExponentsIndependent Set 3.01 **ANSWER KEY**

Six questions, on your own, in silence. Show the step that does the work.

1. Write as one power. $3^4 \cdot 3^5$
 3^9

2. Find the value. $2^3 \cdot 2^2$
 $2^5 = 32$

3. Write as one power. $(7^3)^2$
 7^6
Power of a power: multiply.

4. Find the value. $(3 \cdot 4)^2$
 144
 $3^2 \cdot 4^2 = 9 \cdot 16$, or 12^2 .

5. Find the value. $(-2)^3 \cdot (-2)^2$
 $(-2)^5 = -32$
An odd power of a negative base is negative.

6. Find the value of n . $5^2 \cdot 5^n = 5^7$
 $n = 5$
 $2 + n = 7$.